

Robustness Analysis of Machine Learning Models Under Distribution Shift

COMPARATIVE EVALUATION USING

- Cluster Removal
- Feature-Region Removal
- PCA Space Removal

DATASET

UCI Credit Card Default

MODELS EVALUATED

- Logistic Regression
- Decision Tree
- Random Forest
- KNN
- Gradient Boosting
- XGBoost
- LightGBM

Why Standard Metrics Are Not Enough

Traditional Evaluation

- Accuracy
- ROC-AUC
- F1-score

*Assumption: Training & test data share the same distribution
— violated in the real world.*

Real-World Violations

- Unseen customer populations
- Missing feature regions
- Changing behavioral patterns
- Latent feature shifts

*Goal: Evaluate model behaviour under population shift,
missing decision regions & latent-space gaps.*

Three Distribution Shift Experiments

01

Cluster Removal

Removed a subgroup of samples from training

Semantic Shift

02

Feature-Region Removal

Removed 40–60th percentile of top features

Decision-Region Shift

03

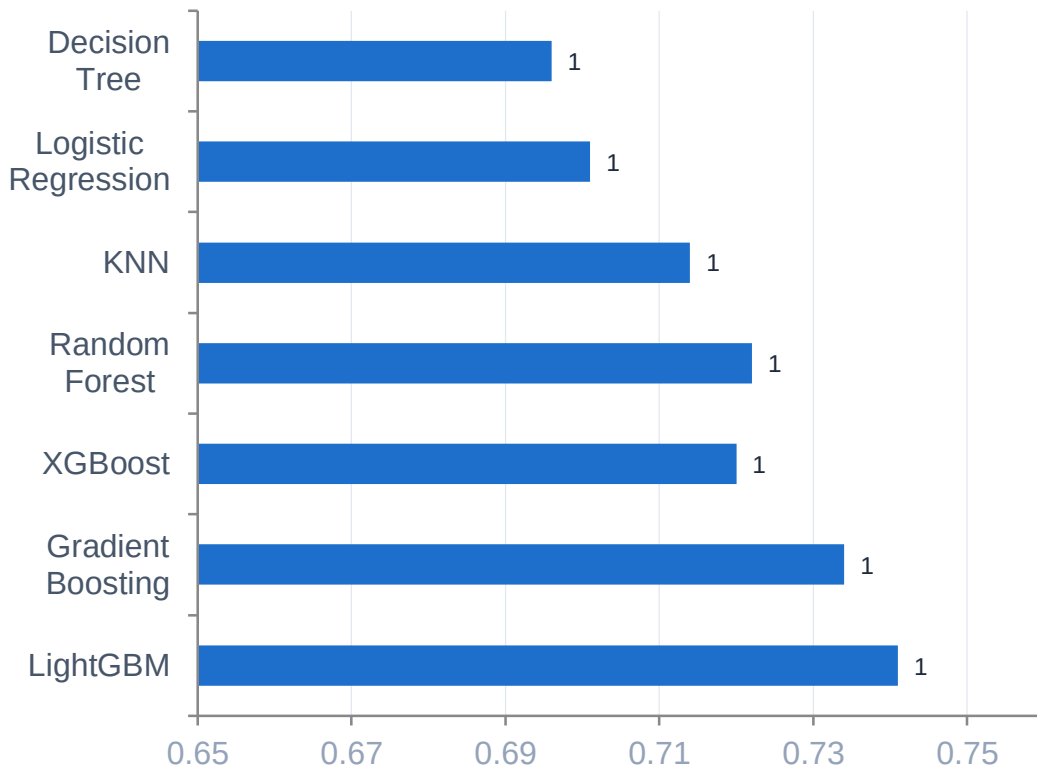
PCA Square Removal

Removed a rectangular region in PCA space

Structural Latent-Space Shift

Generalization → Unseen populations | Interpolation Robustness → Missing regions | Calibration → Confidence reliability

In-Distribution (ID) Accuracy



~5%

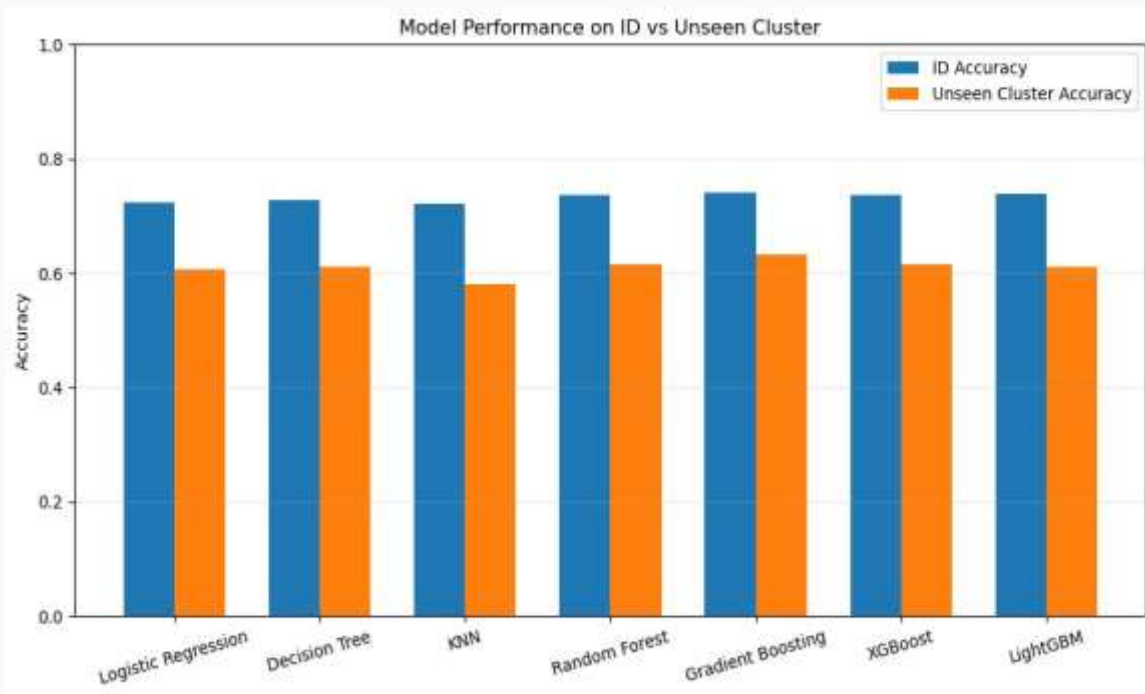
Accuracy spread
across all 7 models

Key Insight

Standard metrics suggest all models are nearly equivalent.

But robustness experiments reveal large hidden differences under distribution shift.

ID vs Unseen Cluster Accuracy



Best: Gradient Boosting

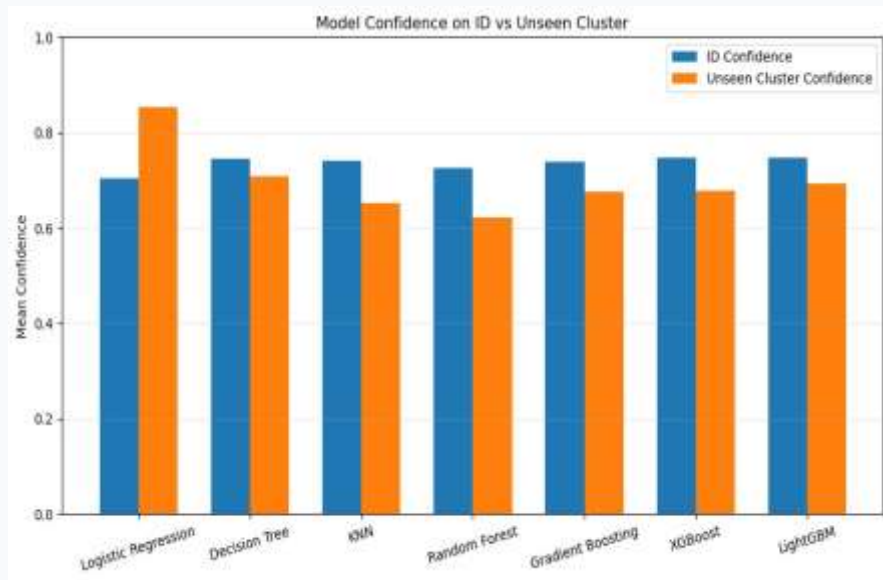
Smallest drop $\downarrow 0.108$
Best interp accuracy: 0.632

Worst: KNN

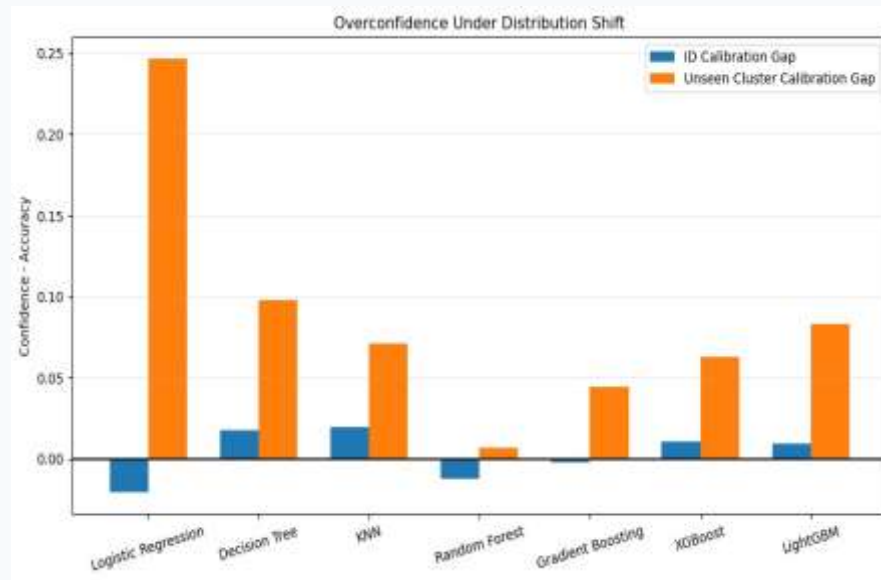
Largest drop $\downarrow 0.140$
Local neighborhoods destroyed

Average drop across all models: $\sim 12\%$ — Removed cluster was not fully decision-critical; models still extrapolate via nearby structure

Overconfidence Under Cluster Shift



Model Confidence



Calibration Gap (Confidence - Accuracy)

Logistic Regression calibration gap: +0.246 — Highly confident but significantly wrong.
This 'confidently wrong' failure mode is dangerous in production systems.

Performance & Confidence: ID vs Interpolation Zone

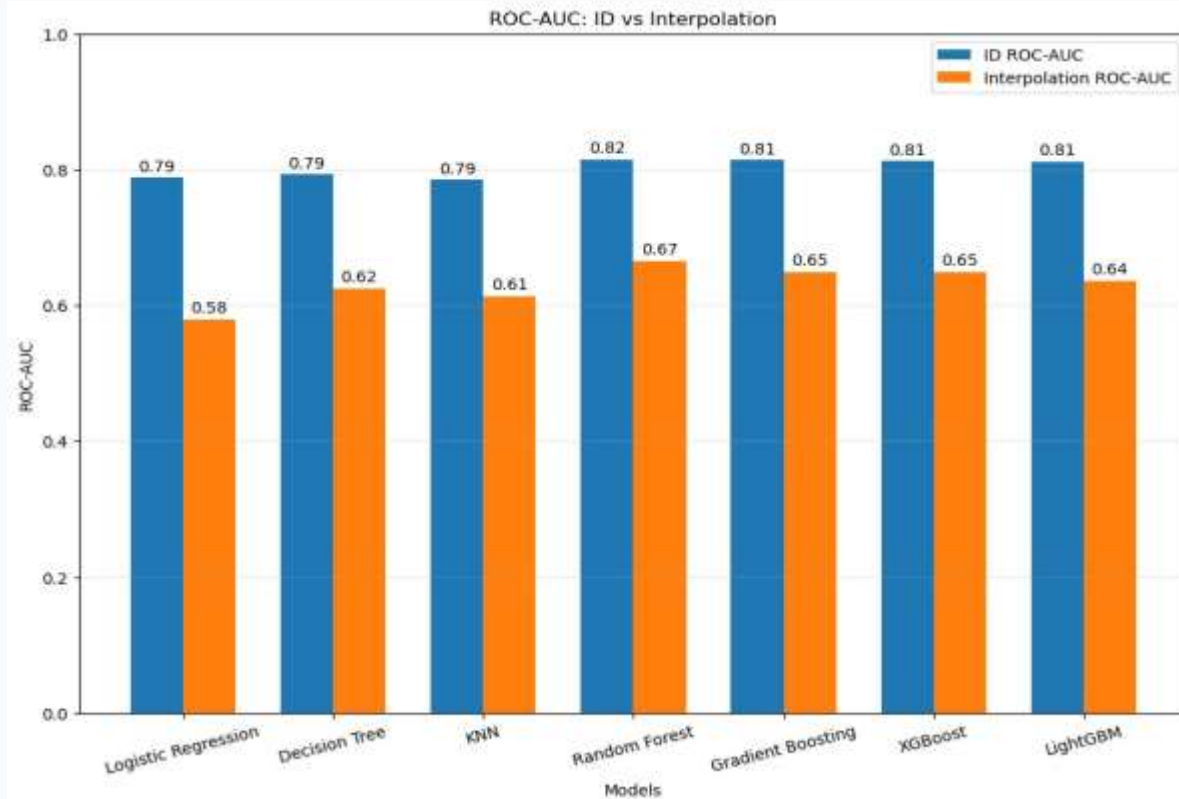


★ Decision Tree ↓0.039

✗ Logistic Regression ↓0.238

✗ KNN ↓0.247

Feature-Region Removal — ID vs Interpolation



Accuracy Spread

Jumped from 0.032 (Exp 1)
to 0.203 (Exp 2)
→ Much harder experiment

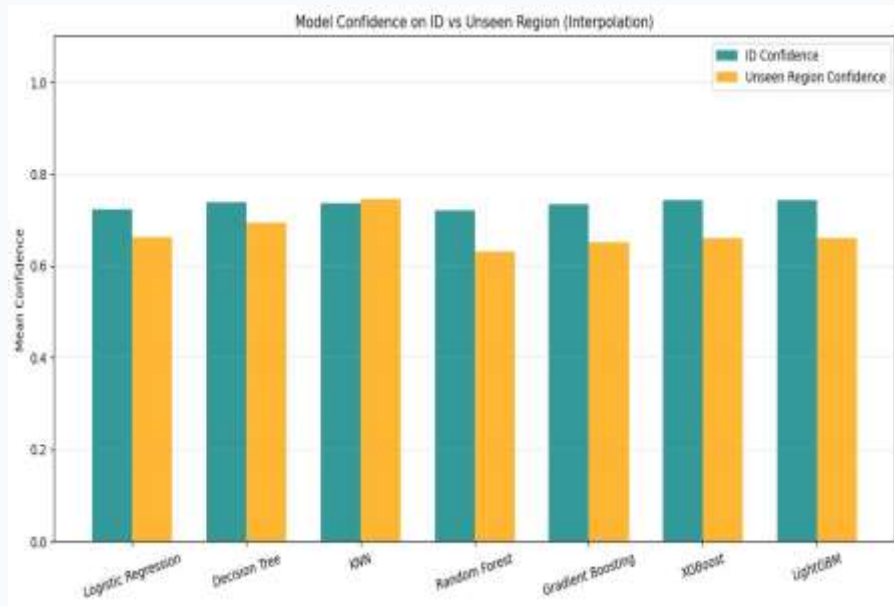
Best ROC-AUC

Random Forest: 0.82 ID
Gradient Boosting stable
across shifts

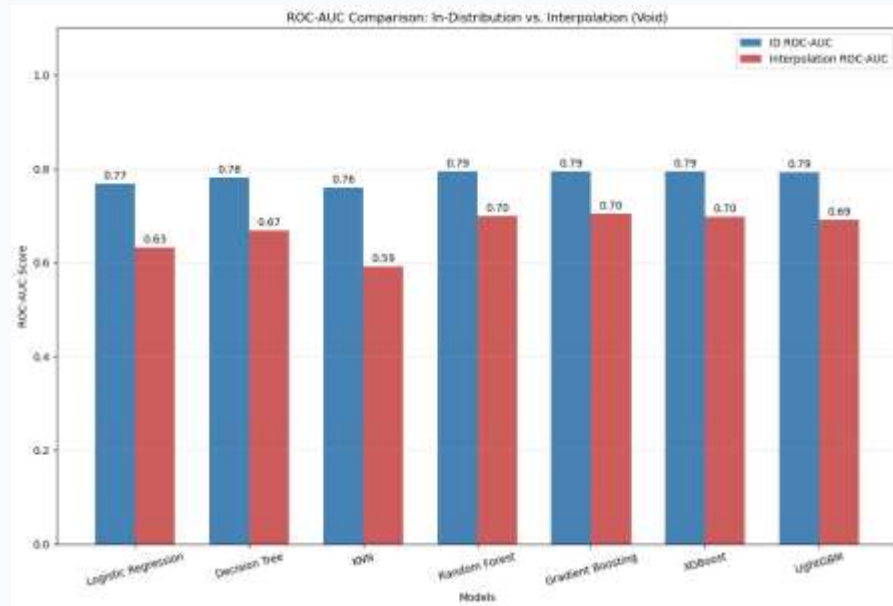
Worst ROC-AUC

Logistic Reg.: 0.58 interp
KNN: 0.61 interp
Both near-random

Confidence & ROC-AUC: ID vs PCA Void



Mean Confidence (ID vs Void)



ROC-AUC (ID vs Interpolation Void)

All models collapse toward ~0.50 accuracy — equivalent to random guessing.
The PCA void removed joint covariance structure; no model retains meaningful support in this latent region.

Models Remain Confident as Accuracy Collapses

Experiment	Avg Accuracy Drop	Avg Confidence Drop
Cluster Removal	~12%	~6%
Feature Removal	~14%	~4%
PCA Removal	~16%	~0%

MOST DANGEROUS CALIBRATION GAPS

Decision Tree	+0.485
KNN	+0.293
Logistic Reg.	+0.246

As shift severity increases, accuracy drops sharply while confidence barely changes.

*Core Insight: Confidence scores measure boundary certainty — NOT training data support.
Models can be simultaneously highly confident and significantly wrong under distribution shift.
This is critical in healthcare, finance, and autonomous systems.*

Final Model Ranking Under Distribution Shift

1	Gradient Boosting	Best robustness + calibration
2	LightGBM	Strong and stable
3	XGBoost	Robust but sharper boundary fitting
4	Decision Tree	Surprisingly stable in Exp 2
5	Random Forest	Moderate robustness
6	Logistic Regression	Severe extrapolation failure
7	KNN	Worst under missing regions

Core Experimental Findings

01 IID accuracy hides robustness failures

Models with similar normal-test accuracy behaved very differently under distribution shift.

02 Shift type determines failure mode

Cluster, feature-region, and PCA shifts each trigger different failure mechanisms.

03 KNN & Logistic Regression highly vulnerable

Both became confidently wrong and unstable under interpolation tasks.

04 Boosted ensembles are most reliable

Gradient Boosting consistently delivered the best robustness and calibration.

05 Confidence scores cannot be trusted alone

High confidence + low accuracy under shift is critical for healthcare, finance, and autonomous systems.

Next Steps & Improvements

Uncertainty-Aware Methods

- Bayesian models
- Conformal prediction
- OOD detection

Additional Shift Types

- Temporal drift
- Adversarial shifts
- Real-world deployment drift

Improved Reliability Metrics

- Expected Calibration Error (ECE)
- Brier Score
- Predictive entropy

Thank You

Questions & Discussion

"Robustness under distribution shift is more important than standard test accuracy."